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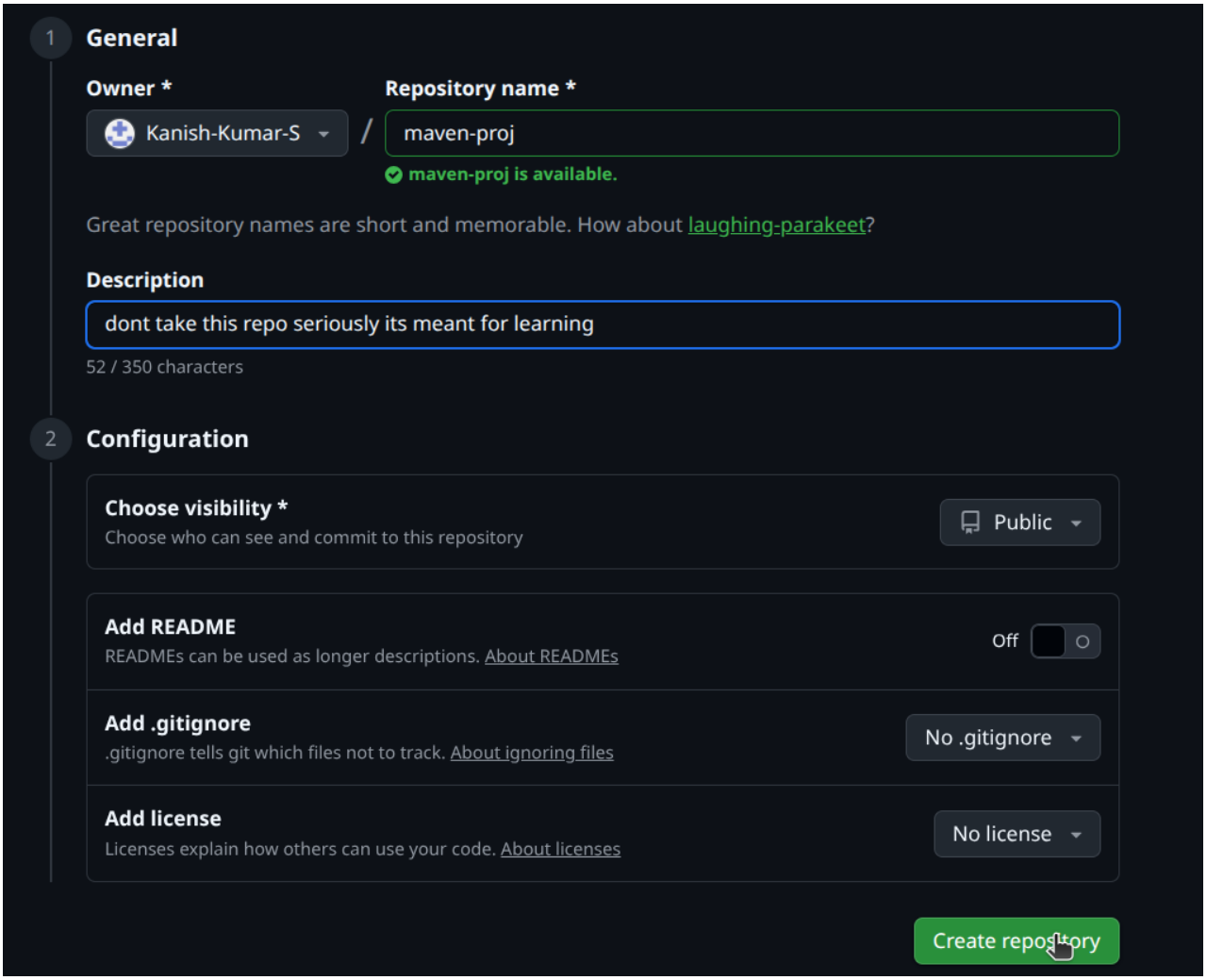
Agile development and devOps lab

DA 3

1. Create a Maven Project in Git Create a new Git repository
   Create a simple Maven project structure manually, including the following files and folders: pom.xml src/main/java/com/example/App.java src/test/java/com/example/AppTest.java
2. Implement the Java Program Write a simple Java program in App.java under your newly created Git repository for any one of the following scenarios: Student Grade Calculator Bank Account Management System Employee Salary Calculator Library Book Management System Simple Calculator Application Write a corresponding unit test in AppTest.java.
3. Push and Clone the Repository . Clone the repository using Visual Studio Code. Verify the project structure.
4. Build the Maven Project Use Maven commands in Visual Studio Code terminal to: Compile the project Run tests Build the project (mvn clean install)
5. Create and Run Jenkins Pipeline Create a new Pipeline project in Jenkins. Create a pipeline script Execute the pipeline and verify successful build completion.


1
General
Owner *
Repository name *
Kanish-Kumar-S
/
maven-proj
maven-proj is available.
Great repository names are short and memorable. How about laughing-parakeet?
Description
dont take this repo seriously its meant for learning
52 / 350 characters
2
Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
Off
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Create repository

```
● ~/Public/student-calculator$git status
  On branch master

  No commits yet

  Untracked files:
    (use "git add <file>..." to include in what will be committed)
          pom.xml
          src/

  nothing added to commit but untracked files present (use "git add" to track)
● ~/Public/student-calculator$git add .
● ~/Public/student-calculator$git add .
● ~/Public/student-calculator$git commit -m "first  commit"
  [master (root-commit) 55edc12] first  commit
   3 files changed, 89 insertions(+)
   create mode 100644 pom.xml
   create mode 100644 src/main/java/com/example/App.java
   create mode 100644 src/test/java/com/example/AppTest.java
● ~/Public/student-calculator$git remote add origin https://github.com/Kanish-Kumar-S/maven-proj
● ~/Public/student-calculator$git push origin master
  Enumerating objects: 14, done.
  Counting objects: 100% (14/14), done.
  Delta compression using up to 16 threads
  Compressing objects: 100% (6/6), done.
  Writing objects: 100% (14/14), 1.45 KiB | 1.45 MiB/s, done.
  Total 14 (delta 0), reused 0 (delta 0), pack-reused 0
  To https://github.com/Kanish-Kumar-S/maven-proj
   * [new branch]      master -> master
○ ~/Public/student-calculator$
```

```
● ~/Public/student-calculator$mvn compile
  [INFO] Scanning for projects...
  [INFO]
  [INFO] -----------------------< com.example:calculator >-----------------------
  [INFO] Building calculator 1.0-SNAPSHOT
  [INFO] --------------------------------[ jar ]---------------------------------
  [INFO]
  [INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ calculator ---
  [WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
  [INFO] skip non existing resourceDirectory /home/kanish-kumar/Public/student-calculator/src/main/resources
  [INFO]
  [INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ calculator ---
  [INFO] Nothing to compile - all classes are up to date
  [INFO] ------------------------------------------------------------------------
```

mvn test

```
t-platform-launcher/1.10.2/junit-platform-launcher-1.10.2.jar (184 kB at 806 kB/s)
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.098 s -- in
com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  14.320 s
[INFO] Finished at: 2026-03-09T21:50:14+05:30
[INFO] ------------------------------------------------------------------------
```

mvn clean install

```
[INFO] --- maven-install-plugin:2.4:install (default-install) @ calculator ---
[INFO] Installing /home/kanish-kumar/Public/student-calculator/target/calculator-1.0-SNAPSHOT.jar to /home/kanish-kumar/.m2/repository/com/example/calculator/1.0-SNAPSHOT/calculator
-1.0-SNAPSHOT.jar
[INFO] Installing /home/kanish-kumar/Public/student-calculator/pom.xml to /home/kanish-kumar/.m2/repository/com/example/calculator/1.0-SNAPSHOT/calculator-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.151 s
[INFO] Finished at: 2026-03-09T22:01:48+05:30
[INFO] ------------------------------------------------------------------------
```

EXPLORER

STUDENT-CALCULATOR
- src
  - main/java/com/...
    - App.java 2
  - test/java/com/e...
    - AppTest.java 2
- target
- pom.xml M

App.java 2 | AppTest.java 2 | pom.xml M

src > main > java > com > example > App.java > App

```java
package com.example;

public class App {

    public int add(int a, int b){
        return a + b;
    }

    public int subtract(int a, int b){
        return a - b;
    }

    public int multiply(int a, int b){
        return a * b;
    }

    public int divide(int a, int b){
        if(b == 0)
            throw new ArithmeticException("Division by zero");
        return a / b;
    }

    Run | Debug
    public static void main(String[] args){
        App calc = new App();
        System.out.println("Addition: " + calc.add(a: 10,b: 5));
    }
}
```

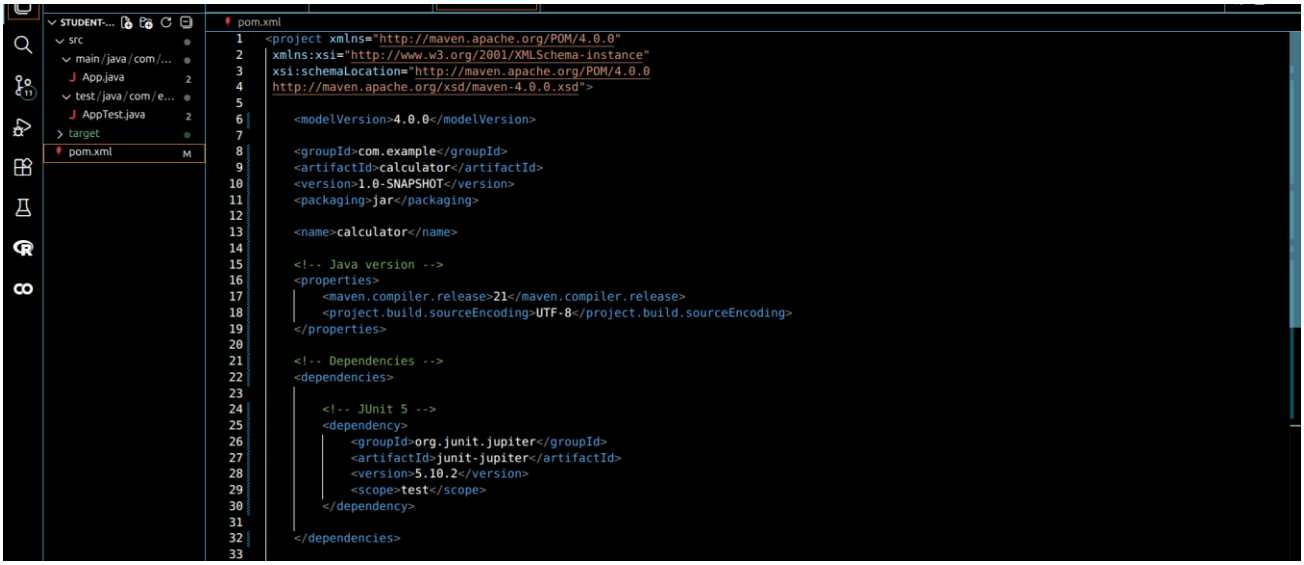
EXPLORER

STUDENT-CALCULATOR
- src
  - main/java/com/...
    - App.java 2
  - test/java/com/e...
    - AppTest.java 2
- target
- pom.xml M

App.java 2 | AppTest.java 2 | pom.xml M

src > test > java > com > example > AppTest.java > AppTest > testAdd()

```java
package com.example;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;

public class AppTest {

    App calc = new App();

    @Test
    public void testAdd(){
        assertEquals(5, calc.add(2,3));
    }

    @Test
    public void testSubtract(){
        assertEquals(1, calc.subtract(3,2));
    }

    @Test
    public void testMultiply(){
        assertEquals(6, calc.multiply(2,3));
    }

    @Test
    public void testDivide(){
        assertEquals(2, calc.divide(6,3));
    }
}
```




```
~/Public/student-calculator$touch Jenkinsfile
~/Public/student-calculator$git add Jenkinsfile
~/Public/student-calculator$git commit -m "added jenkins file"
[master 8354857] added jenkins file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 Jenkinsfile
~/Public/student-calculator$git push origin master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 324 bytes | 324.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Kanish-Kumar-S/maven-proj
   55edc12..8354857  master -> master
~/Public/student-calculator$
```

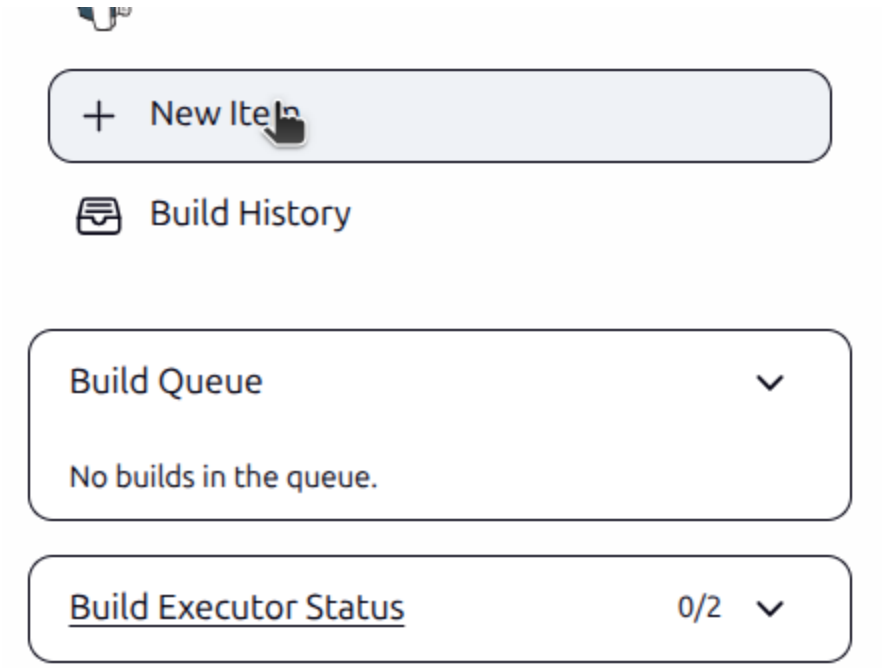
App.java 2 | AppTest.java 2 | Jenkinsfile

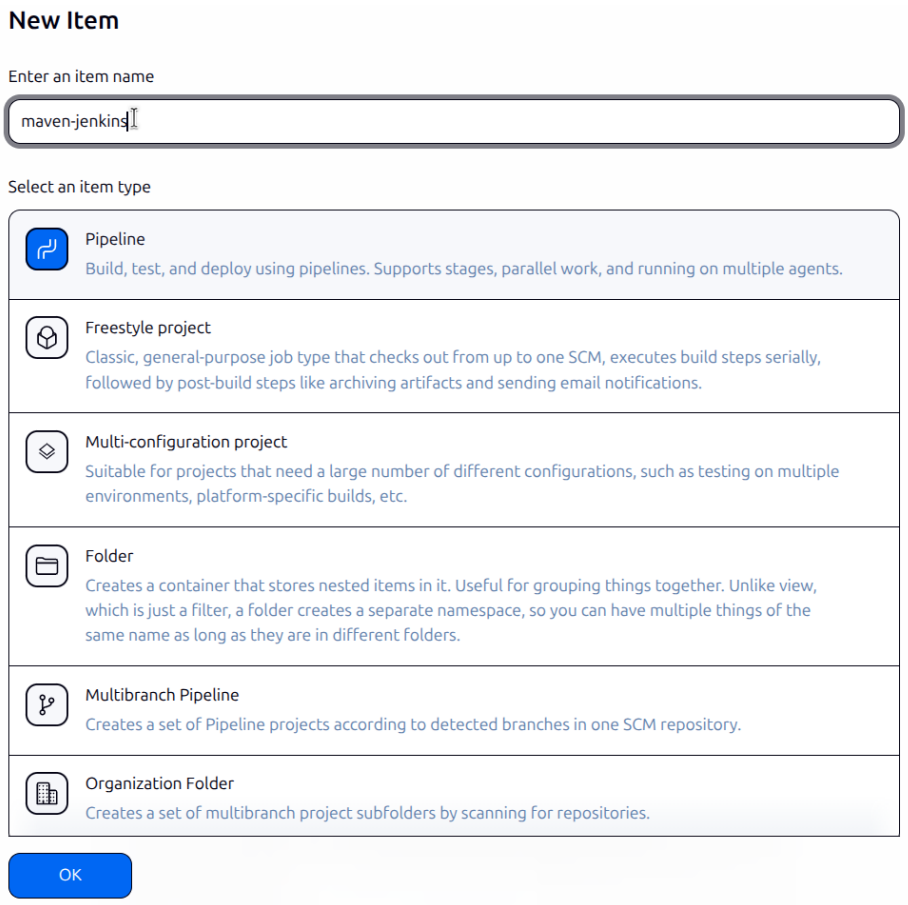
Jenkinsfile

```
pipeline {
    agent any

    stages {

        stage('Clone') {
            steps {
                git 'https://github.com/Kanish-Kumar-S/maven-proj.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean compile'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }

        stage('Package') {
            steps {
                sh 'mvn clean package'
            }
        }
    }
}
```

PROBLEMS 4 | OUTPUT | DEBUG CONSOLE | PORTS | TERMINAL | bash

```
~/Public/student-calculator$git add .
~/Public/student-calculator$git commit -m "4 commit"
[master 28c803d] 4 commit
 1 file changed, 4 insertions(+), 10 deletions(-)
~/Public/student-calculator$git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 437 bytes | 437.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Kanish-Kumar-S/maven-proj
   27fe4a0..28c803d  master -> master
~/Public/student-calculator$
```


New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2


New Item
Enter an item name
maven-jenkins
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
OK

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/Kanish-Kumar-S/maven-proj

Credentials

- none -

+ Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/master

+ Add Branch

Save

Apply

```
Downloaded from central: https://repo.maven.apache.org/maven2/commons-lang/commons-lang/2.1/commons-lang-2.1.jar (208 kB at 404 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/2.1/plexus-archiver-2.1.jar (184 kB at 356 kB/s)
[[1;34mINFO[m] Building jar: /var/lib/jenkins/workspace/maven-jenkins/target/calculator-1.0-SNAPSHOT.jar
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
[[1;34mINFO[m] [1;32mBUILD SUCCESS[m
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
[[1;34mINFO[m] Total time:  5.329 s
[[1;34mINFO[m] Finished at: 2026-03-09T22:45:30+05:30
[[1;34mINFO[m] [1m------------------------------------------------------------------------[m
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```